



सेंट्रल ट्रान्समिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/19

Date: 09-06-2023

As per distribution list

Subject: 19th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 19th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31st May, 2023 (Wednesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

Kashish
09/06/23
(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 19th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31.05.2023

The 19th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 31/05/2023. List of participants is enclosed at **Annexure-X**.

A. Confirmation on the minutes of the 18th CMETS-NR meeting

It was deliberated that the minutes of the 18th CMETS-NR meeting held on 28/04/2023 were issued vide letter dated 17/05/2023. However, it was informed that in the minutes circulated, based on CTU cost committee inputs, following elements to be incorporated in present scope of agreed transmission scheme for “Modifications in Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex)”:

- 1 no. of BC & TBC at 220kV level of 765/400/220kV Barmer-I PS (shifted from future scope of Barmer-I PS)
- Spare one reactor unit (110MVAR) at Rishabdeo/Salumbar PS for 765 kV Rishabdeo/Salumbar - Mandsaur PS D/c line

Further based on CTU cost committee inputs, 1 no. of additional 400kV bay is to be included in present scope of 400/220kV Jhangi PS (shifted from future scope) for GIS diameter completion under “Transmission system for evacuation of power from Shongtong Karcham HEP (STKHEP) & Tidong HEP” in Himachal Pradesh

Further, no comments were received from any applicant, accordingly, minutes were confirmed as circulated. The detailed agenda was discussed as below:

B. Application related matters in Northern Region

1. Treatment of Applications Granted/Received under Connectivity regulations, 2009 under Transition process of CERC (GNA) Regulations, 2022

- It was deliberated that Hon’ble CERC vide notification dated 01-04-2023 has issued First Amendment of GNA Regulations and notified that the effective date of Regulation 37.1 to 37.8 (ie. Cases for arrangement of Transition) shall be 05-04-2023 in place of 15-10-2022. In line with the regulations, options were needed to be exercised by the concerned entities afresh on the basis of their status as on 05-04-2023 within one month from date of coming into effect of the GNA Regulations i.e. by 04-05-2023.
- It was noted that for such applicants who have opted for GNA regime, the identification of transmission system required for grant of Connectivity under GNA regime is under process. Accordingly, it was deliberated that all such Connectivity applications shall be discussed upon evolution of corresponding transmission system expectedly in next CMETS/separate meeting.

2. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022:

It was deliberated that following 8 Nos. of applications for Connectivity to ISTS were received in the month of April'23 (from 05/04/2023 onwards):

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
1.	2200000003	BN Hybrid Power-1 Private Limited	Barmer distt., Rajasthan	14.04.2023	119.2/ 30.06.2025	Generator (Hybrid: Solar-40MW, Wind-79.2MW))	30.06.2025	Land Route	Fatehgarh-IV PS
2.	2200000024	Sprng Power Private Limited	Barmer distt., Rajasthan	17.04.2023	400/ 31.12.2025	Generator (Solar)	31.12.25	Land BG Route	Fatehgarh-IV PS
3.	2200000027	Renew Surya Vihaan Private Limited (Enhancement)	Jaisalmer distt., Rajasthan	18.04.2023	10/ 31.10.2023	Generator (Solar)	31.10.2023	Land Route	Fatehgarh-III PS
4.	2200000008	ACME Cleantech Solutions Private Limited	Bikaner distt., Rajasthan	18.04.2023	300/ 31.05.2025	Generator (Solar)	31.05.2025	Land Route	Bikaner-III PS
5.	2200000018	Sprng Pavana Urja Private Limited (Enhancement)	Barmer distt., Rajasthan	26.04.2023	100/ 31.12.2025	Generator (Solar)	31.12.2025	Land BG Route	Fatehgarh-IV PS
6.	2200000013	Avaada RJBikaner Private Limited	Bikaner distt., Rajasthan	27.04.2023	50/ 31.03.2025	Generator (Solar)	31.03.2025	Land Route	Bikaner-III PS
7.	2200000020	Serentica Renewables India Private Limited	Barmer distt., Rajasthan	29.04.2023	300/ 31.03.2025	Generator (Solar)	31.03.2025	Land Route	Fatehgarh-IV PS

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
8.	2200000040	#Sprng Akshaya Urja Private Limited	Bikaner distt., Rajasthan	29.04.2023	75/ 30.06.2025	Generator (Solar)	30.06.2025	LOA from REMCL	Bhadla-III PS

It was deliberated during the meeting that Applicant had withdrawn their Connectivity application vide their letter dated 15/05/23 in view of revised RTC Project configuration. Accordingly, this Connectivity application was considered as closed. Applicant agreed for the same during the meeting.

Further for Connectivity applications indicated at **Table-1**, it was deliberated that the optimal transmission system requirement for above applications could only be assessed after completion of GNA transition of applicants granted connectivity under 2009 regulation. In view of this, the above applications shall be discussed in the CMETS/Separate meeting with detailed transmission system required for grant of Connectivity in line with GNA Regulations subsequently.

3. Drawl of 200 MW power by M/s Hindustan Zinc Limited (Bulk Consumer) at Kankroli 400/220 kV ISTS Substation in Rajasthan

It was deliberated that Connectivity was earlier granted to M/s Hindustan Zinc Limited as a Bulk Consumer during the 14th NR-CMETS meeting held on 23/12/2022. In line with the discussions during the meeting, Connectivity was granted through the following transmission system:

- HZL– Kankroli (PG) 220 kV D/c line along with associated line bays at both end (under the scope of applicant) [Line capacity shall be suitable to carry minimum 300 MW per circuit at nominal voltage]

Further, it was also mentioned that M/s Serentica vide their letter dated 17/05/2023 informed that they have signed up Power Delivery Agreement (PDA) with Hindustan Zinc Limited (HZL) to deliver Round the Clock Renewable Energy power for 200 MW. For the project, Wind is proposed to be set up at Kallam, Maharashtra while Solar will be set up at Bikaner, Rajasthan.

Earlier, during the 17th NR-CMETS meeting held on 31/03/2023, the two nos. of LTA applications from M/s Serentica Renewables India 4 Private Limited, for 100 MW each, with injection point as Kallam substation (WR), for which M/s HZL was the drawee entity (as a Bulk Consumer) at Kankroli ISTS 400/220 kV substation were also discussed.

Based on the NRLDC inputs & CTUIL revised studies, considering the peak load of Kankroli complex, loading of Kankroli ICTs are observed to be N-1 non-compliant with drawl of above load (200 MW) and need of 1 No. of 500 MVA ICT (4th) ICT augmentation at Kankroli S/s for such drawl of power emerged. Following Transmission system was agreed for drawl of power of 200 MW:

- 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s along with associated 220kV bay
- Shifting of existing 50 MVar Bus reactor [so as to accommodate ICT] and development of bay for reactor
- Extension of 220 kV side of ICT through 220kV Cable/GIB

Now, during the present meeting, it was deliberated that M/s Serentica has requested vide their letter dated 17/05/2023 that to meet their intended project timelines of July'24 for Solar & December'24 for Wind, a long-term access permission associated with a transmission element i.e. 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s may pose a significant uncertainty on the drawl of power. M/s HZL also mentioned that out of 200 MW, a major portion of the capacity (ie. upto 150 MW-160 MW) can be accommodated with existing infrastructure.

In addition to this, M/s HZL has also opted for GNA transition process as a bulk consumer. Accordingly, as per the GNA Regulations, the transmission system required for drawl of power as a bulk consumer by M/s HZL shall be evolved, which includes the requirement of 1 No. of ICT at Kankroli Substation. In view of the same, M/s Serentica vide above letter had requested to facilitate an optimal solution through which bulk of the capacity of 200 MW can be provided access to the grid without the requirement of ICT.

In view of above, views from Grid India were sought once again and they stated that as discussed earlier, there is nearly 60-70 MW margin available at present in 3*315 MVA Kankroli ICTs. As informed by HZL, presently they are drawing 96 MW from RVPN network and would be shifting their drawl of power from Kankroli RVPN substation to Kankroli PG substation. Since, presently also they are drawing from Kankroli (PG) ICTs via Kankroli RVPN substation, they could possibly drawn early 150MW (96MW+60 MW) from Kankroli (PG) ICTs which could be further checked by CTUIL in their study base case.

In line with this, CTUIL suggested that keeping in view the future load growth, some margin can be kept from planning point of view and power upto 140 MW may be drawn with the existing transmission system as per the present studies. Therefore, for the balance (60 MW) of quantum, 1x500 MVA.400/220kV ICT (4th) at Kankroli S/s shall be required. However, as it was applied for 200 MW under one application, this cannot be considered in split quantum.

Further, it was also suggested that under clause 22.4 'Effectiveness Date of GNA', wherein it has been clarified that "For entities covered under Regulation 17.1, where such entity seeks to advance the date from which GNA shall be effective for full or part quantum of GNA granted, the Nodal Agency shall make such GNA effective, subject to availability of transmission system." Accordingly, M/s HZL may seek effectiveness of GNA for Part quantum in future, based on availability of margin at that time without availability of 4th 400/220 ICT at Kankroli, however, full GNA quantum shall be made effective only with the commissioning of required ICT at Kankroli S/s. Alternately, it was also suggested that they may apply afresh with splitting the total quantum of 200 MW into different quantum so as Connectivity under GNA regime can be granted separately.

M/s Serentica/HZL further enquired whether they can change their quantum of Connectivity opted under GNA transition from 200 MW to 140 MW now. On this CTUIL clarified, since the process of submission of documents under GNA transition (connectivity) was allowed till 04/05/2023, therefore any such request shall require deliberation with CERC also.

M/s Serentica/HZL informed that they shall revert on above discussed options within a week.

Subsequently, M/s Serentica/HZL vide their email dated 08/06/2023 informed that as stated above, out of three options given to them for optimally drawing power from Kankroli S/s, they have decided to go ahead with clause 22.4 of GNA Regulations, wherein M/s HZL may seek effectiveness of GNA for Part quantum in future, based on availability of margin at that time without availability of 4th 400/220 ICT at Kankroli, however, full GNA quantum shall be made effective only with the commissioning of required 1 No. of 400/220 kV ICT at Kankroli (ISTS) S/s.

In view of above, GNA (200 MW) shall be granted to M/s HZL upon completion of GNA Transition process as per the GNA Regulations. Further, HZL may request for effectiveness of GNA for Part quantum in future as per GNA regulation clause 22.4.

C. ISTS Expansion

i. Construction of intrastate downstream transmission system from 400/220 kV, 2X315 MVA Kala Amb (ISTS) Substation of Powergrid Kala Amb Transmission Limited (PKATL) - (Agenda by HPPTCL)

It was deliberated that in order to meet the present and future load requirement of the area, HPPTCL in 31st Standing Committee on Power System Planning of Northern Region held on 2nd January' 2013 had proposed for establishment of a 400/220/132 kV substation at Kala Amb by LILO of one circuit of either N'Jhakri – Abdullapur or Karcham Wangtoo – Abdullapur 400 kV line. Further, as Karcham Wangtoo – Abdullapur 400 kV D/c line was also passing in close proximity to Kala Amb area and to meet the loads in

the area, the LILO of the Karcham Wangtoo – Abdullapur 400 kV D/c was agreed to be carried out at Kala Amb and it was decided that 220 kV and 132 kV works shall be carried out by HPPTCL as per their requirement.

Accordingly, 400/220 kV, 2X315 MVA substation at Kala Amb was established by PKATL under ISTS (TBCB) along with 6 Nos. 220 kV bays as per HPPTCL requirement. HPPTCL is now executing following downstream Intrastate Transmission System by utilizing 4 No. 220 kV bays and the same shall be commissioned within next 3-4 months:

- a. 220 kV (Zebra) D/C line on M/C Towers from 400/220 kV Kala Amb Substation PKATL to 220/132/33 kV Andheri Substation of HPPTCL.
- b. 220/132/33 kV (220/132 kV, 200 MVA (Single Phase Transformer Bank) & 132/33 kV, 2X 31.5 MVA) Substation at Andheri by HPPTCL.
- c. 220 kV (Zebra) D/C line from 400/220 kV Kala Amb Substation of PKATL to 220/132/33 kV Giri Substation of HPSEBL. (Part of line from 400/220 kV Kala Amb PKATL till Devani is on M/C towers)

CTUIL inquired about the total drawl of power through 2 Nos. of 400/220 kV ICTs at Kala Amb. HPPTCL confirmed that about 240 MW of total power shall be drawn through above mentioned 220 kV downstream network, therefore, N-1 criterion shall be complied in this case.

NRLDC representative enquired whether there will be any injection from 220kV side to 400V side during high hydro season at 400/220kV Kala Amb substation given that 220kV line connectivity is planned to 220/132kV Giri substation. HPPTCL representative confirmed that they would be only drawing power as Giri generation would be consumed locally and there would not be a case of injection of power from 220 kV side to 400 kV side.

In view of above, the 220 kV downstream network being interconnected with Kala Amb 400/220 kV Substation as proposed by HPPTCL was noted.

ii. Transmission system strengthening for interconnections of Bhadla-III & Bikaner-III complex

It was deliberated that Comprehensive Transmission scheme for evacuation of power from Rajasthan REZ Ph-IV (Part-1) (Bikaner Complex) is under bidding for power transfer of 7.7GW incl. 4GW from Bikaner-III PS. The scheme comprises 765kV EHVAC corridor from Bikaner-III PS towards load centers of Delhi/UP. About 0.85 GW connectivity application granted/received at Bikaner-

III PS (out of which few applications may be shifted at Bikaner-II PS to utilize vacated margin of about 0.675 GW at Bikaner-II PS in GNA). In view of that about 0.25GW RE applications will be available at Bikaner-III PS against 4GW RE evacuation capacity.

St-II Connectivity for about 3 GW and LTA of about 1GW, against the potential of 6.5 GW (under Ph-III), was received/granted at Bhadla-III PS. Earlier, St-II Connectivity for about 2.75 GW RE and LTA of 2.6GW, against the potential of 2.9 GW (under Ph-III), was received/granted at Ramgarh PS. However, M/s Adani surrendered 2.1 GW connectivity/LTA at Ramgarh PS under GNA, considering above, power transfer requirement from Ramgarh will be 0.65 GW (2.75GW-2.1GW) in GNA regime.

As part of Rajasthan REZ Ph-III (20GW) Transmission scheme, Bhadla-III & Ramgarh PS are being established for integration of 6.5GW & 2.9 GW RE potential respectively. Ramgarh PS is also being inter-connected with Bhadla-III for evacuation of RE power. In order to facilitate evacuation of 9.4GW RE power from Ramgarh/Bhadla-III PS (6.5GW+2.9GW) from Bhadla-III onwards, 765kV Bhadla-III - Sikar-II D/c line with implementation schedule of Dec'24 (Tentative) [for about 2.9GW power transfer requirement] as well as 6GW HVDC corridor (± 800 kV Bhadla (HVDC) -Fatehpur(HVDC)) with implementation schedule of Dec'26 (42 months schedule) is being implemented as part of Ph-III scheme.

As mentioned above, Connectivity for 3.65GW (0.65GW at Ramgarh + 3GW at Bhadla-III) is already granted/under process at Ramgarh/Bhadla-3 PS. In view of expected implementation schedule of HVDC system beyond Bhadla-3 (Dec'26), there is a need of additional corridor from Bhadla-3 onwards beyond 2.9GW RE potential of Ramgarh/Bhadla-3 PS.

Further, at Bikaner-III PS, application for only 0.25 GW is left against 4 GW planned evacuation capacity. Since, more injection is anticipated than planned EHVAC capacity at Bhadla-III, in the requisite time frames (25-26), there is a need for optimal utilization of Bikaner-III corridor capacity by forming a high capacity tie interconnection between Bhadla-III & Bikaner-III PS.

Accordingly, 765kV Bhadla-III - Bikaner-III D/c line is proposed to facilitate optimal utilization of EHVAC corridor beyond Bikaner-III PS for evacuation of RE generation including at Ramgarh/Bhadla-III PS.

Further, as part of committee report "Transmission system for integration of over 500GW capacity by 2030" as well as MNRE/SECI inputs, a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan is evolved comprising 10 GW RE potential (Wind: 4 GW, Solar: 6 GW) along with 3GW BESS(net evacuation 5 GW) at Ramgarh in Jaisalmer complex by 2030. Out of 10 GW potential by 2030, in its Ph-I (by 2025) 3GW potential (Wind: 2GW, Solar: 1GW) was informed at Ramgarh. As part of

above committee report, additional 3 GW RE injection at Ramgarh PS and 2GW RE injection at Bhadla-III/Bhadla-IV PS is envisaged beyond 2025 (Ph-II/III), which will utilize the planned HVDC corridor beyond Bhadla-3 as it will match implementation schedule of generation and transmission i.e. in 2026-27 timeframe.

In the MOP meeting held on 01.05.23, it was deliberated that GIB committee clearance is still pending for 400kV Fatehgarh-II-Bhadla-III D/c line under Ph-III Part-B1 package as above line is emanating from Fatehgarh-II PS, which is falling under GIB core area and adversely impacting progress of other linked packages. Accordingly, it was directed that process of delinking of 400kV Fatehgarh-II- Bhadla-III D/c line from Ph-III Part-B1 may be carried out at the earliest and a separate package may be formed comprising this line. It is understood that RE developers are also facing challenges in getting DTL clearance from GIB committee at Fatehgarh-II PS. Deferment of 400kV Fatehgarh-II- Bhadla-III D/c line will also reduce the SCR at Bhadla-III PS/Bhadla HVDC which is already at boarder line. Therefore, 765kV Bhadla-III PS - Bikaner-III PS D/c line shall also provide additional advantage to improve short circuit strength in Bhadla-III complex including for LCC HVDC terminal at Bhadla.

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-3 :2 GW) (Ramgarh Complex) was deliberated in 17th CMETS-NR meeting held on 31.03.23 which included 765kV Ramgarh-Bhadla-III PS - Bikaner-III PS D/c Corridor. However considering connectivity surrendered at Ramgarh by M/s Adani (2100 MW) the proposal was reviewed as above.

CTU stated that system studies were carried out under various scenarios considering 765 kV Bhadla-III – Bikaner-III D/c tie interconnection and study files were also circulated on 25.05.23. From the study results, it was observed that line loading with proposed strengthening is in order in normal as well as N-1 contingency condition. It is also observed that in Feb solar maximized scenario (revised case), loading of 400kV RAPS-Shujalpur D/c line is marginally higher (about 900MW) in N-1 contingency. Additionally loading of 765/400kV Bareilly ICT (2X1500MVA) is marginally higher in some scenarios (n-1:1550MW). The loading of the above line and ICT will be reviewed with progress of RE generation projects at Rajasthan and strengthening requirement will be identified later, if required. Further, short circuit level at Bikaner-III PS is also within limits (765kV-29.3kA ,400kV- 55kA, 220kV- 24.3 kA) with proposed strengthening.

It was stated that in view of the above considerations, 765 kV Bhadla-III – Bikaner-III D/c tie interconnection shall provide following advantages:

- Optimal utilization of EHVAC transmission system beyond Bikaner-III PS while providing flexibility of power transfer from Bhadla/Bikaner RE clusters

- Additional evacuation requirement from future RE generation of 181.5 GW (Ph-1/2) RE potential at Ramgarh/Bhadla-III (2 GW)
- Meeting evacuation requirement from Bhadla-III PS onwards for same RE generators coming up in 2025-26
- Improves SCR in Bhadla-III complex including for LCC HVDC terminal at Bhadla
- Tie connection will also increase resiliency of transmission corridor in both the complexes i.e. Bhadla & Bikaner

Grid-India stated that studies are in order and above proposal is agreeable to them.

Further, Grid-India in the meeting as well as vide mail dated 31.05.23 informed that multiple transmission line tripping in a short duration have been observed in Rajasthan RE complex during inclement weather condition in recent days. Some of these lines tripped due to tower collapse/damage and are still under long outage. This has resulted in backing down of RE generation also to keep the line loadings within permissible limits in the depleted network. (List of lines under long outage due to tower collapse/damage in Rajasthan RE Complex is as under

- 400 KV Bhadla (RVPN) - Merta (out since 25.05.2023)
- 400 KV Bhadla (RVPN) - Jodhpur (out since 25.05.2023)
- 400 KV Jaisalmer - Barmer Ckt-1 (out since 30.05.2023)
- 400 KV Jaisalmer - Barmer Ckt-2 (out since 30.05.2023)
- 400 KV Bhadla (RVPN) - Bikaner (RVPN) Ckt-1 (out since 15.05.2023)
- 400 KV Bhadla (RVPN) - Bikaner (RVPN) Ckt-2 (out since 15.05.2023)
- 765 KV Phagi (RVPN) - Bhiwani (PG) Ckt-1 (out since 25.05.2023)
- 765 KV Bikaner - Khetri ckt 1 was out on 27.05.2023 due to crossarm damage

It was further mentioned that the tripping of multiple lines during inclement weather especially due to tower collapse/damage is a matter of serious concern and suggested following measures in order to avoid such incidents in future.

- Review of the Wind Zones in Rajasthan RE Complex on priority
- Enhancing resiliency of the transmission system through:
 - Planning of single ckt lines in place of double ckt lines in critical/high loading corridors
 - Applicability of N-2 reliability criteria or beyond may be considered on case to case basis in natural disaster/inclement weather prone areas. Historical data of multiple element outage in such areas and feedback of system operator may be referred in this regard while planning of the transmission system.

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- It was agreed that input will be deliberated in a separate meeting and the final recommendations will be put up in subsequent NCT for approval.

It was agreed that input will be deliberated in a separate meeting and the final recommendations will be put up in subsequent NCT for approval. CTU stated that the matter regarding review of wind zone and resiliency of the transmission system was also deliberated in the 66th NRPC meeting wherein it was decided that a committee shall be formed to deliberate on issue & mitigating measures

NLDC representative further informed that, in the study cases, a large quantum of power (**~9200 MW**) is getting evacuated from 765/400 kV Bikaner PG making it one of the most critical station in the complex. Pooling of such large quantum of power at a single station is not desirable from resiliency point of view as complete station outage may result in cascade tripping. He further requested CTUIL to explore alternatives paths with suitable network rearrangements so as to reduce large evacuation of power from a single station.

CTUIL stated that RE pooling stations in western Rajasthan complex (Bikaner/Bhadla/Fatehgarh) are contagious to each other and power from above complexes shall flow towards load centres of Northern & Western region mainly through following high capacity corridors

- For Punjab (765kV Moga) and Delhi/NCR (765kV Jhatikara, Narela, Gurgaon) load centres through 765kV Bikaner and Khetri & Neemrana-II (via Bikaner-III)
- To UP load centres (Aligarh, Varanasi, Kanpur, Bareilly & Balia) through 765kV Bhadla-II/III (via 765kV Sikar-II), 400kV Bhiwadi (via 400kV Bikaner-II), Fatehpur HVDC (via Bhadla-III HVDC) & Neemrana-II (via Bikaner-III)
- To Western Region load centres (765kV Gwalior & 765kV Mandsaur) through 765kV Beawar/Dausa (via Fatehgarh-III) & Sirohi/Rishabdeo (via Fatehgarh-IV/Barmer-I)

In view of integration of more than 50GW in ISTS system in western Rajasthan, most of RE pooling station i.e. Bikaner (PG) may be utilized for evacuation of about 7-9 GW RE generation (incl self RE generation of 4-6GW) towards load centers. To make the system more resilient, additional transmission system may be required which may increase the transmission system investment (up to 1.5 times) for evacuation of the RE power mainly through solar without BESS which is available only in day time. In view of this, detail deliberations are required on above matter on resilience of transmission system for RE vis-à-vis transmission system additional cost and optimal utilization of transmission system as above requirement is beyond transmission planning criteria.

NLDC representative further highlighted that below mentioned lines are heavily loaded in the study case shared by CTUIL. The angular separation in the base case and under N-1 is also on the higher side. Same may also be addressed through suitable rearrangements as mentioned above.

S. No.	Transmission Line	Loading (MW)	Angular Separation (Base Case)	Angular Separation (under N-1)
1.	765 kV Bhadla – III – Sikar D/C	2x1760	19 deg.	25 deg.
2.	765 kV Bikaner – Moga D/C	2x1963	20 deg.	25 deg.
3.	765 kV Bikaner-III – Neemrana 2xD/C	4x1868	15 deg.	19 deg.

CTU stated that as per Transmission Planning Criteria, 2023, angular separation between adjacent buses under 'N-1' shall not exceed 30 degree. Further it is mentioned that in transmission planning efforts to be made that angular difference between adjacent buses may not exceed 20 degree in base case and 30 degree in N-1 contingency (except Inter regional and critical high capacity links). Additional transmission system/ rearrangements may be required to further reduce the angular difference between adjacent buses which may increase the transmission system investment.

Further, in the 8th CMETS-NR meeting held on 30.06.22, result of P-V, Q-V analysis for such highly loaded 765kV lines was depicted by CTU as part of minutes of above meeting. From the result it was observed that the system was stable under various contingencies.

- 765kV Bhadla-3-Sikar-2 D/C (350km)
- 765kV Sikar-2-Aligarh D/c (275km)
- 765kV Bikaner-Khetri D/c (240km)
- 765kV Bikaner-Moga D/c (366km)
- 765kV Khetri-Jhatikara D/c (146km)
- 765kV Bhadla-2-Sikar-2 D/c (310km)

CEA & Grid-India also concurred on above proposal and no comments were received from other stakeholders. In view of deliberation held in present CMETS-NR meeting, following ISTS Transmission scheme was agreed:

- 765 kV Bhadla-III – Bikaner-III D/c line along with 240 MVar switchable line reactor for each circuit at Bhadla-III end (~150 km)

Estimated Cost : Rs 1382 Cr

Note : 765/400kV ICTs , 400/220kV ICTs and 220 kV line bays (for RE connectivity) at Bhadla-III shall be taken up separately based on RE development/applications at Bhadla-III PS

However, on the matter of resiliency of the transmission system raised by Grid-India, it was decided to have deliberation in a separate meeting after NRPC committee meeting as discussed above.

List of Participants of 19th Consultation meeting for Evolving Transmission Schemes in NR held on 31.05.2023

CEA

Shri Nitin Deswal	Deputy Director
Shri Kanhaiya Singh Kushwaha	Assistant Director

SECI

Shri R.K.Agarwal	Consultant
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Grid India

Shri Alok kumar	GM
Shri Gaurav Malviya	Manager
Shri Priyam Jain	Manager
Shri Gaurab Dash	Assistant Manager

NRPC

Shri Rajat Dixit	AEE
Shri Santosh Kumar	SE

CTU

Shri Jasbir Singh	CGM (CTU)
Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Madhusudan Meena	Engineer (CTU)

HVPNL

Shri Sanjay Arora CE (PD&C)

HPPTCL

Shri Rajneesh Kumar GM (C&D)

Shri Harmanjeet Singh Assistant Engineer

LTA/Connectivity Applicants

Shri Ajay Kumar Pradhan	Acme Cleantech Solutions Private Limited
Shri Rajat Sangwan	Acme Cleantech Solutions Private Limited
Shri Yogesh Kumar Sanklecha	Acme Cleantech Solutions Private Limited
Shri Pradeep	Acme Cleantech Solutions Private Limited
Shri Angshuman Rudra	Avaada Rjbikaner Private Limited
Shri Ashish Shukla	Avaada Rjbikaner Private Limited
Shri Vineet Taneja	BN Hybrid Power-1 Private Limited
Shri Mohit Jain	ReNew Surya Vihaan Private Limited
Shri Arzaan Dordi	Serentica Renewables India Private Limited
Shri Vishnu Khandelwal	HZL
Smt Poorva Pitke	Sprng Power Private Limited
Shri Sumit Joge	Sprng Power Private Limited